Image Data Augmentation and Image Classification

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Foundation course / Section -1

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1.Abstract

This project focuses on Image data Augumentaion and image classification. By using basic image preprocessing and augmentation methods using Opencv. Basic Image trasformations such as shifting , resizing and rotaion . These trasformaions help in uderstanding image manipulation techniques.Image augmentation techniques such as resizing and random horizontal flipping are applied using PyTorch.This project helps in understanding how image data augmentation improves model training and supports image classification.

2.Introduction

This project is mainly about Image data Augmentation and Image Classification, which comes under image processing and machine learning.Image classification means identifying and grouping images into different categories/ clusters . In real life problems , we do not have enough images to train a model properly , so image data augmentation is used to solve this problem.

In this project image data augmentation techniques are applied to increase the number of training images.The augmentation techniques such as like rotating , flipping,resizing, and shifting images are used to create new images from existing ones. For classification purposes , the cat and Dog dataset was used .This helps in model to learn better and improves the accuracy of image classification.

The project implemented using python programming with important libraries such as NumPy , torch , torchvision and Opencv . PyTorch and Torchvision libraries were used to build a transformation pipeline and load the dataset for training. Basic machine learning concepts are also used to classify images.The procedure includes loading images,applying augmentation,training the model, and check the results. The main purpose of this project is to understand how image data augmentation improves the image classification performace.

Topics that i received during intership training:

- Basics of python like Data , varibles,lists,loops,Data structures, Functions.
- OOPS concepts like class ,object,

  polymorphism,Abstraction,Encapsulation, Inheritance
- Important libraries like NumPy and Pandas
- Breif overview on machine learning
- Regression and Classification
- LLM Fundamentals
- Communication skills

3.Project Objective

The objectives of this project are:

- To understand the basic image preprocessing and image classification.
- To study how augmented images help improve the accuracy of image classification models.
- To compare the model performace before and after applying image data augmentation.
- To load and mange image datasets using pytorch

4.Methodology

The project Image Data Augmentation and Image Classification was carried out in a structured manner involving image preprocessing , dataset preparation ,augmentaion, and data loading for deep learning tasks.

Step – 1: Importing Required Libraries

The necessary python libraries were imported such as

- Numpy for numerical operations
- OpenCV (cv2) for image processing
- matplotlib for show the images
- Torch and Torchvision for deep learning and dataset handling

These libraries provided the tools required for image manipulation and classification preparation.

Step – 2: Image Loading

The image was downoaded and loaded using "cv2.imread()".

The shape of the image was examined using the ".shape" function to understand its dimensiond (height,width,color channels).

This step ensured that the image was properly loaaded into the working environment.

Step – 3: Image Preprocessing

several image transformation techniques were appiled using OpenCV:

1. Grayscale conversion

The original image was converted into grayscale by using cv2.cvtcolor().

2.Saving processed Image

The grayscale image was saved using cv2.imwrite().

3.Image Shifting

An affine transformation matrix was created to shift the image 50 pixel to the right and 100 pixel downward using cv2.warpAffine().

4.Image Resizing

The image was resized to 150 * 100 pixel using cv2.resize() to study simension scaling.

5.Image Rotation

The image wasr rotated 90 degrees using cv2.getRotationMatrix2D() and cv2,warpSffine().

These preprocessinf operations helped in understanding spatial transformations and their effect on image data.

Step – 4: Dataset Collection and preparation

The cat and Dog dataset was downloaded from the provided link using python’s zipfile module.

The dataset directory structure was mainted properly so that it could be used with torchvision‘s ImageFolder class.

Step – 5: Image Data Augmentation

To improve datset variblity a transormation pipeline was created using torchvision.trandforms.Compose() . Includes random horizontal flipping, resizing image to 255 *255 and coversion of image into Tensor format.

Step – 6: Dataset creation using ImageFolder

The training dataset was created using datasets.ImageFolder().

This will automatically assigns labels based on folder names.

Total number of images in the dataset was verified to ensure proper loading.

Step – 7: DataLoader Creation

A data loader was created using torch.utils.data.DataLoader().

Parameters batch size:64 ,shuffle : True.

The DataLoader enlabled batch processing of images and labels, which is essential for training deep learning models efficiently.

Tools and Technologies Used

- Python
- Numpy

- OpenCV
- matplotlib
- pyTorch
- Torchvision
- Google colab

5.Data Analysis and Result

5.1 Descriptive Analysis

The original image was processes using various opencv transformation tecniques. The dimensional changes observed are summarized below.

Table -1 : Image Transformation summary

Operation	Output Shape	Description
Original image	(800,640,3)	RGB image with 3 color channels
Grayscale image	(800,640)	Single channel intensity image
Shifted image	(800,640,3)	Image without size change
Resized image	(100,150,3)	Image scaled to smaller resolution
Rotated image	(640,800,3)	Height and width swapped after rotation

Observations

- Grayscale conversion reduced color channels from 3 to 1.
- shifting preserved original dimensions.

- Resizing reduced computational complexity.
- Rotation altered spatial orientation while maintaing image information.

5. 1. 2 Dataset Analysis

The cat and dog dataset was downloaded and loaded sucessfully using ImageFolder.

Table – 2: Dataset Summary

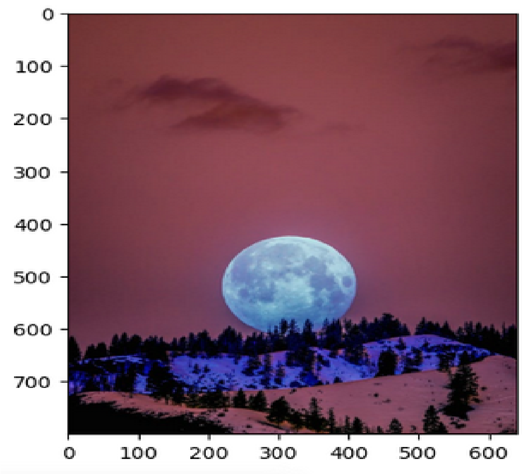
Parameters	Values
Total Training Images	22,500
Number of classes	2 (Cat and Dog)
Resized Dimensions	255 * 255
Batch size	64
Image tensor shape	(64 , 3 , 255 , 255)

Observstions

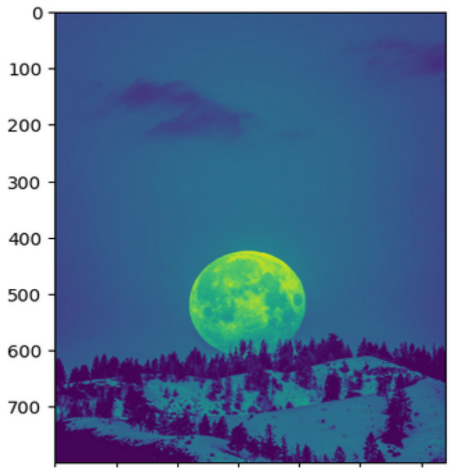
- The dataset contains 22,500 images in total.
- Images were resized to 255 *255 pixels.
- DataLoader sucessfully generated mini- batches of 64 images.

5. 2 Visualization Results

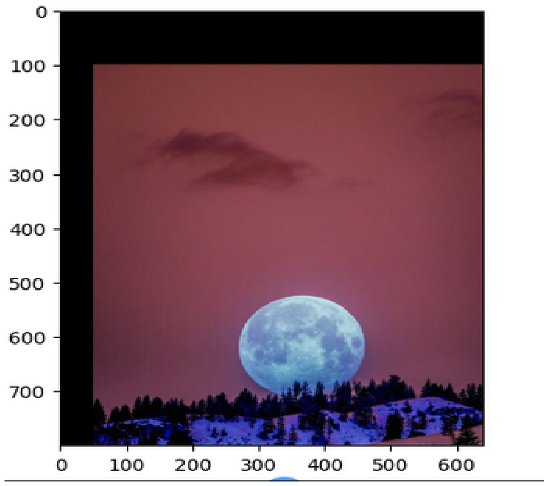
1.Original image

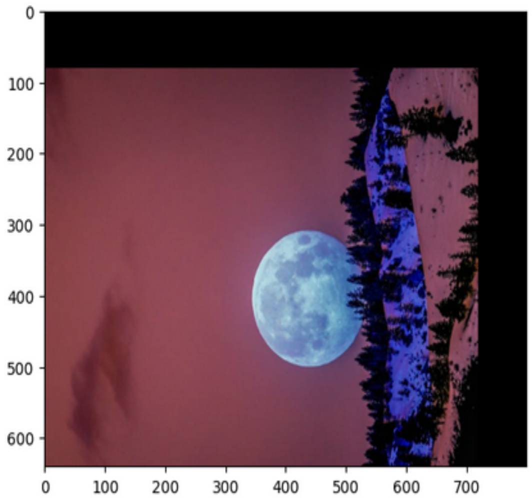


2.Grayscale image

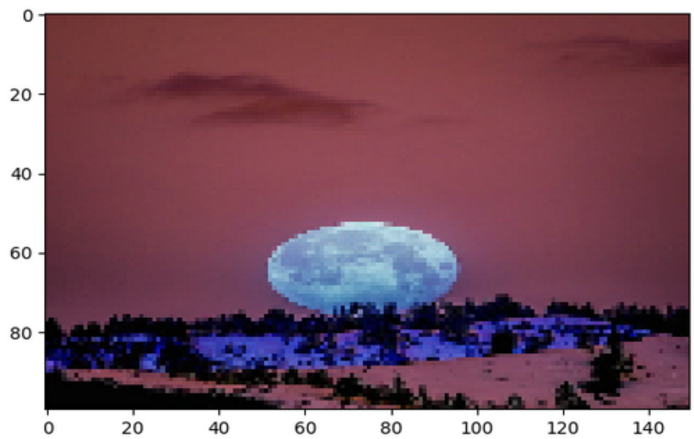


3. shifted image





4.Rotated image



5. Resized image

5.3 Inferential Analysis

In this phase of the project , no statistical hypothesis testing or inferential stastical analysis was performed.

The focus of the project was on :

- Image preprocessing
- Data augmentaion
- Dataset preparation for classification.

5 . 4 Machine learning Model Analysis

In this phase of the project ,no machine learning model was developed or trained. The primary focus was on image preprocessing , data augmentaion, and dataset preparation for image classification .

6. Conclusion

The project sucessfully demonstrated image preprocessing and data augmentaion techniques for image classification tasks.

From the results:

- image transformation such as grayscale conersion , shifting, resizing and rotation were sucessfully implemented using opencv.
- The cat and dog dataset containg 22,500 training images was properly downloaded ,extracted and structured.
- A trasformaion pipeline was created using Pytorch and Torchvision.

This findings show that proper preprocessing and augmentaion are essential step before training deep learning models. The prepared dataset is now suitable for building advanced image classification models.

Recommendations for future work:

- Implement and train a CNN-based image classification model using the augmented dataset.
- Evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.
- Apply advanced augmentation techniques such as random zoom, noise injection, and contrast enhancement.

7. APPENDICES

Appendix 1: References

- opencv offical Documentaion
- PyTorch offical Documentaion
- Torchvision Documentaion
- Cat and Dog Dataset

Appendix 2 : Github link

Github link : https://github.com/usharanipadi-8520/Image-Data-Augmentation-and-Image-Classification

Appendix 3 : Documentaion Links

Datset link : https://drive.google.com/drive/folders/1TeLp4U4NsXCSgClbF7ODBsaLKpHSWeQr?usp=sharing

Google colab link :

https://colab.research.google.com/drive/1HhCR4yHtSDJFLFPahdJSTqzrmO-q_G_D?usp=sharing